

Question Paper

Exam Date & Time: 01-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Responsible and Safe AI Systems [NPT_4133]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Summarize the major AI capability improvements over the past 5-10 years. How do these improvements increase AI risk exposure? (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

B) Demonstrate how goal misspecification can affect the performance and outcomes of AI systems in real-world applications such as recommender systems or self-driving cars. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Explain the principles of Transparency and Accountability in AI and demonstrate how applying them can help prevent AI misuse. (5)
A)

Course Outcome: "CO2" / Cognitive Level: "Understand"

B) Examine how adversarial attacks affect superhuman game-playing agents such as AlphaGo and analyze their implications for AI safety. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 3) Analyze how mechanistic interpretability contributes to understanding the internal workings of deep learning models, and evaluate the challenges or limitations that constrain its effectiveness. (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

B) Illustrate how data poisoning and Trojan attacks undermine the reliability of AI systems by examining their underlying mechanisms and the ways they distort model behaviour. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Critically evaluate various bias measurement techniques and assess their effectiveness and applicability within AI auditing frameworks. Justify which techniques provide the most reliable foundation for ensuring fairness and accountability in AI systems. (5)
A)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

B) Assess the global Responsible AI regulatory landscape with special reference to India and the EU. (5)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 5) Critically evaluate the ethical implications of AI in the healthcare sector, focusing on issues of transparency and accountability. Assess the impact of failures in these areas through relevant examples, and justify what ethical frameworks or measures could best address them. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- B) The pursuit of AGI could pose risks of power-seeking and deception." - Critically examine this statement. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4133:Responsible and Safe AI Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	5
3	3	CO 3	15
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4133:Responsible and Safe AI Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	1	5(10%)
4	Analysis	3	15(30%)
5	Create	4	20(40%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)